1️⃣ Class & Object (Java OOP)

🔹 Class কি?

Class হলো blueprint / template যেটা দিয়ে object তৈরি করা হয়।

সহজ ভাষায়
Class = Design
Object = Real Product

🧠 Real Life Example

ধরো:

Mobile Phone

Mobile বানানোর আগে একটি **design** থাকে।

Design → Class
Actual Phone → Object

Class : Mobile
Properties : color, brand, price
Behaviour : call(), message()

🔹 Class Structure

```
class ClassName{
  variables (data)
  methods (functions)
}
```

🔹 Example Class

```
class Student {
  String name;
  int id;
  void display(){
    System.out.println(name + " " + id);
  }
}
```

এখানে

Student → class
name,id → variable
display() → method

🔹 Object কি?

Class থেকে তৈরি হওয়া **instance** কে Object বলে।

Student s1 = new Student();

এখানে

Student → class
s1 → object
new → memory allocate

🔹 Real Life Example

ধরো একটি **Car class**

class Car{
color()
brand()
speed()
}

এখন object:

Car c1 = new Car();
Car c2 = new Car();

c1 → BMW
c2 → Toyota

- **Complete Example**

```
class Car {

  String brand;
  int price;

  void show(){
    System.out.println(brand + " " + price);
  }

}

public class Main {

  public static void main(String[] args) {

    Car c1 = new Car();

    c1.brand = "BMW";
    c1.price = 5000000;

    c1.show();

  }
}
```

- **Output**

```
BMW 5000000
```

- **Dry Run**

Step by step execution

Step 1

Program start

main() run

Step 2

Car c1 = new Car();

Memory create

Heap Memory
c1
brand = null
price = 0

Step 3

c1.brand = "BMW"
c1.price = 5000000

Memory

brand = BMW
price = 5000000

Step 4

c1.show()

Print

BMW 5000000

- **Memory Diagram**

```
Stack          Heap

c1 ---------->  Car Object
                brand = BMW
                price = 5000000
```